

The COR Lipid Registry: the effectiveness of the strike early and strike strong strategy in patients with high cardiovascular risk

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Aims

According to real-world data, only up to 20% of patients with atherosclerotic cardiovascular disease (ASCVD) are below the recommended LDL-cholesterol (LDL-C) target of <55 mg/dL (<1.4 mmol/L) 1–3 months after the index event. Accordingly, improved strategies for initiating lipid-lowering therapy (LLT) are desired to achieve treatment targets required to further reduce future cardiovascular event rates.

Methods and results

The COR Lipid Registry included patients at very high cardiovascular (CV) risk presenting for percutaneous coronary intervention (PCI) with acute coronary syndrome (ACS) or chronic coronary syndrome (CCS). Coronary artery disease (CAD) patients with an LDL-C level of >130 mg/dL (equalizing 3.37 mmol/L) or a non-HDL of 160 mg/dL (equalizing 4.14 mmol/L) either LLT-naïve or with suboptimal LLT at index hospitalization were enrolled. Based on lipid levels at baseline, these patients were assumed to need triple LLT to achieve their LDL-C target. Baseline characteristics and lipid parameters of all patients were collected at index hospitalization and two follow-up visits, after 4–6 and 8–12 weeks, respectively. Initially, in all patients a dual LLT (high-dose, highly effective statin, which means atorvastatin 40 mg or 80 mg and rosuvastatin 20 mg or 40 mg, plus ezetimibe) was initiated during the index hospitalization, before PCSK9 inhibitors or bempedoic acid were added, if LDL-C target levels were not met at control visits. In total, 165 very high-risk patients were included, of which 79 (42.0%) were admitted for CCS and 109 (58%) for ACS, respectively. At Visit 1, 114 (69.1%) patients reached the recommended (ESC/EAS guidelines 2019) lipid goals (LDL-C of <55 mg/dL; equalizing 1.4 mmol/L; or non-HDL-C of 85 mg/dL equalizing 2.2 mmol/L), while 160 patients (97%) met the target at Visit 2. In ACS patients, 74.3% reached the recommended treatment goal (LDL-C or non-HDL-C) at Visit 1 and 97.3% at Visit 2. In CCS patients, the treatment goal was achieved in 60.8% at Visit 1 and 100% at Visit 2, respectively. In the LLT-naïve group at admission ($n = 64$), the treatment goal was met in 61.5% of patients at Visit 1 and in 96.9% of patients at Visit 2, while for LLT-pre-treated patients, the respective data were 76.3% and 100%, respectively.

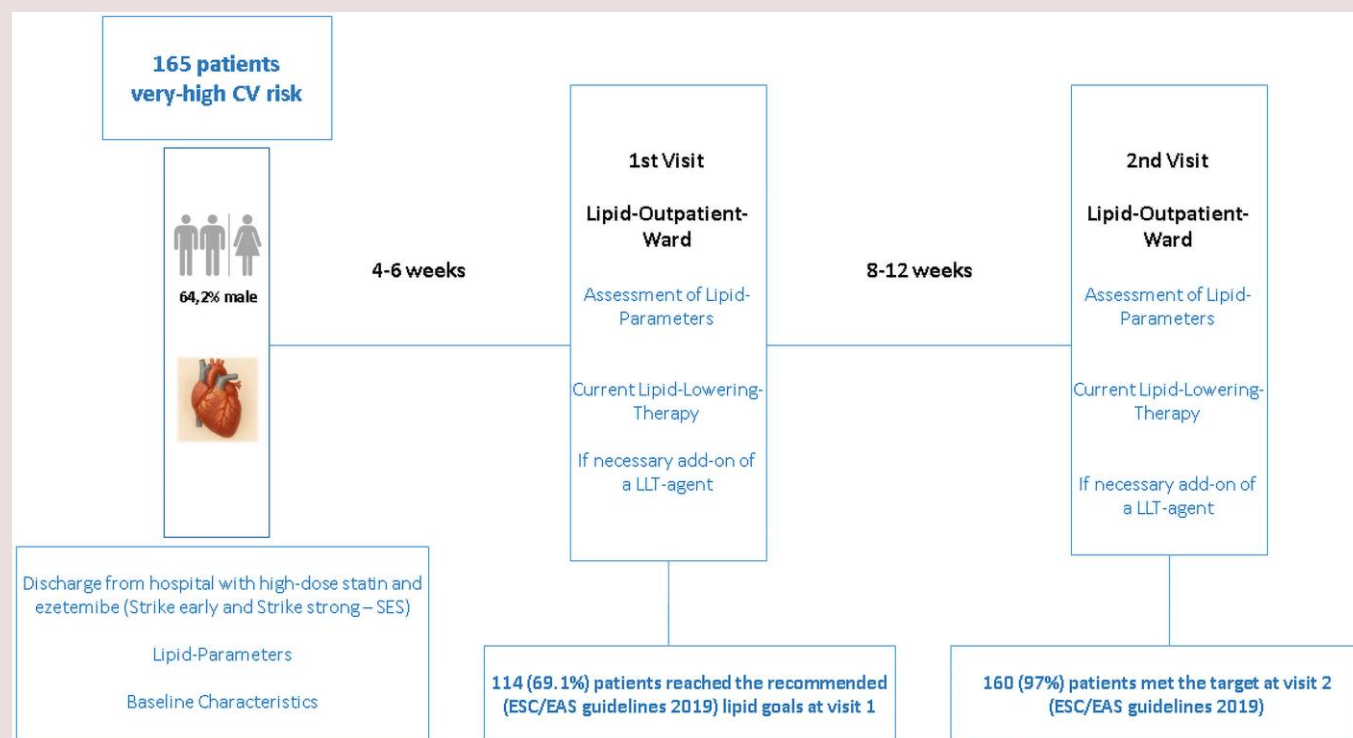
Conclusion

An optimized LLT strategy for managing a very high CV risk patient cohort undergoing PCI, starting with a dual LLT strategy consisting of a high-dose, highly effective statin plus ezetimibe at the index event followed by addition of a third LL-agent (a PCSK9 inhibitor or bempedoic acid) at follow-up after 4–6 weeks if treatment goal was not reached, was highly effective in achieving LDL-C and non-HDL-C goals. Wider adoption of this strategy may help to significantly improve LDL-C target levels in real-world populations in very high-risk CAD patients presenting with ACS or CCS for coronary revascularization.

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Graphical Abstract



Keywords

ACS • CCS • Hyperlipidemia • Prevention • Atorvastatine • Rosuvastatine

Introduction

Cardiovascular disease (CVD) is the leading cause of death worldwide and a major contributor to disability, posing a substantial financial burden on healthcare systems.¹ Effective strategies to mitigate the impact of CVD focus on managing causal risk factors such as arterial hypertension, type 2 diabetes, and hyperlipidaemia, respectively. Furthermore, promoting lifestyle modifications such as smoking cessation, increased physical activity, and weight reduction can significantly improve individual health outcomes.^{1,2}

According to the 2019 ESC/EAS Guidelines for the Management of Dyslipidaemias,³ the 2023 ESC Guidelines for Management of the Acute Coronary Syndrome,⁴ and the 2025 Focused Update of the 2019 ESC/EAS Guidelines for the management of dyslipidaemias,⁵ a stepwise approach remains a Class I recommendation in the management of hypercholesterolaemia. This approach involves the escalation of lipid-lowering therapies (LLT) every 4–6 weeks, starting with high-potency statins, followed by the addition of ezetimibe, bempedoic acid, or PCSK9 inhibitors (alirocumab, evolocumab, inclisiran) in order to achieve the target LDL-cholesterol (LDL-C) level in very high CV risk patients of <55 mg/dL (equalizing 1.4 mmol/L) and a LDL-C reduction of at least 50% of the baseline level. If a 2nd major vascular event occurs within 2 years under optimal LDL-C-targeting therapy, an even more stringent LDL-C target of <40 mg/dL (equalizing 1 mmol/L) should be considered.^{3–5}

In contrast to current guidelines, accumulating scientific evidence supports an initial dual LLT for effective management of hypercholesterolaemia, as initial statin monotherapy is likely inferior in reducing future cardiovascular risk. The IMPROVE-IT trial first demonstrated the

superiority of combining statins with ezetimibe compared with statin monotherapy after acute coronary syndrome (ACS).⁶

Despite the availability of highly effective LLT, many patients fail to achieve recommended lipid targets in real-world settings, as shown by the SANTORINI and the DA VINCI studies.^{7,8} SANTORINI, a prospective observational study conducted between March 2020 and February 2021, focused on the highest-risk population and revealed that among patients with evidence of atherosclerotic CVD, only 20.7% reached their recommended target of 55 mg/dL.⁷ DA-VINCI, a retrospective study conducted between June 2017 and November 2018, assessing the LDL-C goal achievement based on the 2016 ESC/EAS guideline and the 2019 ESC/EAS guideline recommendations among 5888 patients also suggested a need for improved LDL-C-lowering strategies. Among the study population, only 54% of the patients reached the recommended 2016 target values of 70 mg/dL, and only about 33% reached the equivalent goals of the stricter 2019 guidelines.⁸ These findings underscore the substantial gap between guideline recommendations and real-world practice, highlighting the urgent need for more effective implementation of LLT strategies.^{7–9} The strike early (start intrahospital before discharge) and strong (immediately beginning with a dual LLT including a high-potency statin and ezetimibe) strategy (SES), initially suggested by the association of acute cardiovascular care of the ESC (ACVC/ESC), is gaining traction^{10–13} and might be a promising strategy to more effectively reach LDL-C targets.¹⁴

This study aimed to evaluate the effectiveness of an intensified, structured, and stepwise approach to hypercholesterolaemia management in very high-risk CVD patients treated at a tertiary hospital.

The approach focused on achieving ESC-recommended LDL-C targets through intensified LLT as well as respective non-HDL-C targets.

Materials and methods

This was a prospective observational cohort study of consecutive patients managed in the lipid outpatient clinic at the cardiology department of Clinic Ottakring in Vienna (COR Lipid Registry), a routine secondary prevention service for patients with coronary artery disease and dyslipidaemia with elevated LDL-C. It aimed to assess the proportion of patients achieving LDL-C recommended by the 2019 ESC/EAS guidelines, but also non-HDL-C target levels, within the framework of a structured follow-up programme, which can be easily implemented in clinical routine.

Study population

Dual LLT, consisting of a high-dose high-effective statin (defined as atorvastatin 40 or 80 mg and rosuvastatin 20 or 40 mg) and ezetimibe, was initiated in all eligible patients during hospitalization at the Cardiology Department of Clinic Ottakring (formerly Wilhelminenhospital).

Patients meeting the following criteria were then invited and recruited at the outpatient lipid clinic approximately 4 to 6 weeks after discharge:

- LLT-naïve very high CV risk patients presenting with ACS or CCS undergoing PCI and an LDL-C of >130 mg/dL
- Very high CV risk patients presenting with ACS or CCS undergoing PCI on suboptimal LLT pre-treatment, above their recommended ESC target goal

Follow-up visits were scheduled after 4–6 weeks to monitor lipid levels and adjust therapy as needed until the patient achieved the LDL-C goal of <55 mg/dL (1.4 mmol/L) or a non-HDL-C goal of <85 mg/dL (2.2 mmol/L) and at least 50% LDL-C reduction in accordance with the ESC/EAS 2019 guidelines. The baseline visit was performed during index hospitalization, and follow-up Visit 1 (FU1) was conducted 4–6 weeks after discharge. If necessary, treatment optimization was already performed at index hospitalization and controlled in the outpatient ward for LLT. The decision for further therapy intensification (e.g. triple therapy) was made at the discretion of the treating physician and FU1 (4–6 weeks after index hospitalization).

Patients were categorized by clinical presentation at the index hospitalization. Acute coronary syndrome was defined as an admission for an ACS (STEMI, NSTEMI, or unstable angina) requiring urgent or emergent management; these patients were routinely scheduled for a structured follow-up visit at discharge. Chronic coronary syndrome was defined as patients who were initially admitted electively for PCI.

Adverse events

Hepatic safety was defined as ALT or AST increase $\geq 3\times$ the upper limit of normal (ULN) confirmed on repeat testing. Creatine kinase (CK) reference range was 0–170 U/L. Creatine kinase elevations were adjudicated individually, considering baseline values and symptoms; a CK-related adverse event was recorded only when accompanied by muscle symptoms or when it led to dose modification or temporary discontinuation. In general, statin intolerance was measured based on patients' anamnesis/self-reporting in form of myalgia as well as abdominal discomfort. Furthermore, rhabdomyolysis was a clinical diagnosis.

Data collection

This study was designed as a prospective, single-centre analysis and conducted at the Clinic Ottakring (COR Lipid Registry), including patients who visited our specialized outpatient ward for LLT optimization between August 2020 and October 2024. The primary aim of the registry

was to evaluate the achievement of the cholesterol goal in very high-risk CVD patients in a structured real-world clinical setting. Baseline characteristics, laboratory parameters, discharge medication, medical history, and follow-up data were collected for each patient and obtained from patients' medical records by trained study personnel. Data monitoring and validation were performed to ensure data quality and completeness, including cross-checking for implausible values and addressing missing data through rigorous data control procedures.

The study protocol adhered to the ethical standards of the Helsinki Declaration and was evaluated and approved by the institutional review boards of the city of Vienna (Nr: EK-25-076-VK).

Statistics

Continuous variables are presented as mean $\pm$ standard deviation (SD) or as median [interquartile range (IQR)], depending on data distribution. Normality was assessed visually using boxplots and statistically with the Kolmogorov–Smirnov test. Differences between normally distributed continuous variables were evaluated using the Student's *t*-test, while the Mann–Whitney *U* test was employed for non-normally distributed variables. Categorical variables are expressed as percentages and were compared using χ^2 tests or Fisher's exact tests, as appropriate. For categorical variables presented in tables, we report *n/N* (%), where *N* is the number of patients with available data for that variable; denominators therefore vary across variables due to missing data (no imputation for descriptive summaries). Denominators are shown explicitly in the tables. To evaluate significant changes in lipid profiles over time, we compared cholesterol values at different time points within the same patient [at baseline, FU1, and 2nd follow-up visit (FU2)] using the non-parametric Friedman test. This method was chosen due to the non-normal distribution of the data and to account for the correlation between repeated measurements within the same patient and assess whether significant changes occurred during treatment. In addition, we analysed the proportion of patients who achieved the LDL-C target after each follow-up visit at the lipidology clinic. Percentages were calculated based only on patients who attended the respective follow-up visit. Patients lost to follow-up were excluded from the efficacy analysis. For statistical analyses, a two-sided *P* ≤ 0.05 indicated a statistically significant difference between groups. Data analyses were performed using IBM SPSS Statistics, version 26.0 (Armonk, NY: IBM Corp), to ensure rigorous and reproducible results.

Results

Baseline characteristics

A total of 257 patients were eligible to participate in this trial. After excluding patients who rejected participation or were not reachable, 217 patients were invited to our outpatient lipid clinic for optimizing LLT after hospitalization for ACS or CCS with a need for coronary revascularization. Of the 217 invited patients, 52 (24%) did not attend the first visit and were therefore excluded from further analyses (Figure 2). Four patients were lost to follow-up during the FU2, and five patients were lost to follow-up during the third follow-up visit (FU3).

The mean age ($\pm$ standard deviation, SD) of the participants was 60 ± 11.1 years, and the majority were male (*n* = 106, 64.2%, Table 1). A history of cerebrovascular disease was recorded in 16 patients (9.7%), while 10 patients (6.1%) suffered from peripheral artery disease. Arterial hypertension was present in 132 patients (80%) and the diagnosis of type 2 diabetes mellitus in 38 patients (23.0%). Self-reported statin intolerance was present in 31 (18.8%) of cases (Table 1).

Cholesterol goal achievement

At outpatient FU1, 104 patients (63.0%) achieved the LDL-C target recommended by the 2019 ESC/EAS guidelines. At FU2 (8–12 weeks

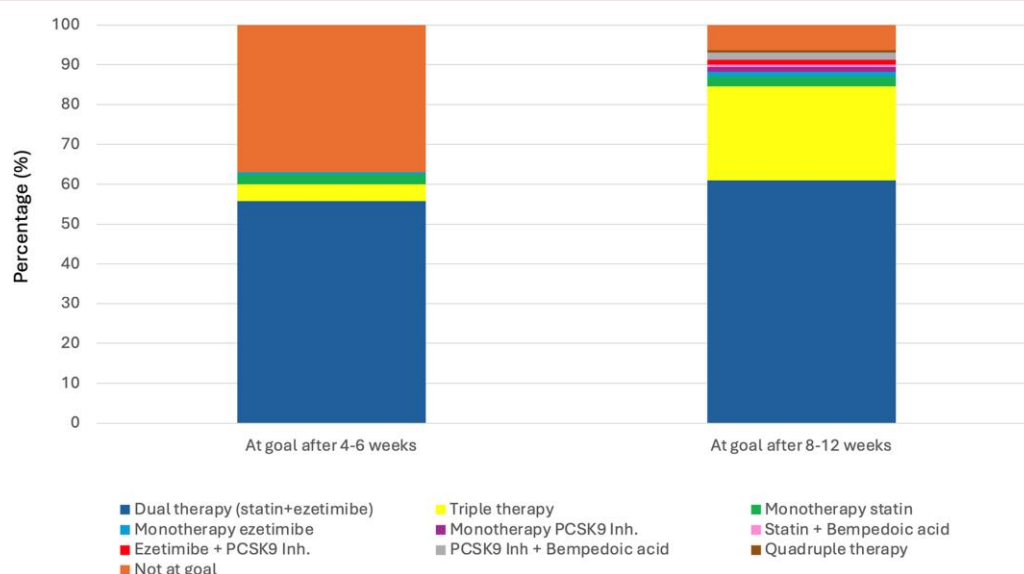


Figure 1 Treatment combinations needed for goal achievement after 4–6 weeks and 8–12 weeks. Dual therapy = statin + ezetimibe (blue), triple therapy (add on bempedoic acid or a PCSK9 inhibitor (yellow), monotherapy statin (green), monotherapy ezetimibe (light blue), monotherapy PCSK9 inhibitors (purple), statin + bempedoic acid (pink), ezetimibe + PCSK9 inhibitor (red), PCSK9 inhibitor + bempedoic acid (grey), quadruple therapy (brown), and not at goal (orange).

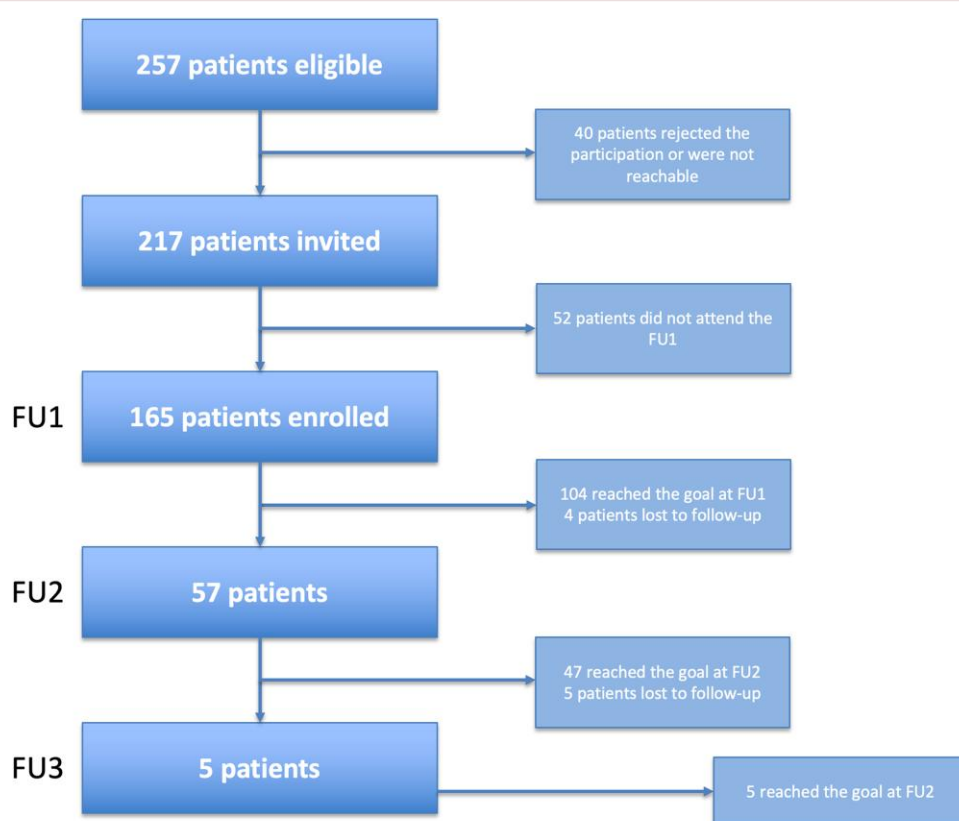


Figure 2 Enrolment flowchart. FU1, follow-up Visit 1 (4–6 weeks after discharge); FU2, follow-up Visit 2 (8–12 weeks after discharge); FU3, follow-up Visit 3 (12–16 weeks after discharge).

Table 1 Baseline characteristics based on the LDL-C target achievement

	Study population (n = 165)	LDL-C goal achieved (n = 104)	LDL-C goal not achieved (n = 61)	P-value
Age (mean, $\pm$ SD)	60 ($\pm$ 11.1)	59 ($\pm$ 10.7)	60 ($\pm$ 11.7)	ns
Male (n, %)	106 (64.2)	68 (65.3)	38 (62.3)	ns
BMI (median, IQR)	28.3 (5.8)	27.4 (6.1)	27.5 (6.3)	ns
ACS prior follow-up in the lipid outpatient clinic, n (%)	109 (66.1)	72 (69.2)	37 (60.7)	ns
Cerebrovascular disease, n (%)	16 (9.7)	6 (5.8)	10 (16.4)	0.026
PAD, n (%)	10 (6.1)	7 (6.7)	3 (4.9)	ns
Arterial hypertension, n (%)	132 (80.0)	81 (77.9)	51 (83.6)	ns
Diabetes mellitus type 2, n (%)	38 (23.0)	23 (22.1)	15 (24.6)	ns
Chronic kidney disease, n (%)	19/163 (11.6)	10 (8.8)	9/59 (15.3)	ns
COPD, n (%)	11/162 (6.8)	6 (5.8)	5/59 (8.5)	ns
Smoking, n (%)	52/110 (47.3)	56 (49.1)	16/49 (32.7)	ns
LLT naive at admission, n/N (%)				
Statin intolerance, n (%)	31 (18.8)	9 (8.7)	22 (36.1)	<0.001
LDL-C at admission (mg/dL) (mean, $\pm$ SD)	146.3 (63.1)	145.0 (44.6)	147.4 (57.9)	ns
Non-HDL-C at admission (mg/dL) (mean, $\pm$ SD)	175.3 ($\pm$ 51.8)	173.5 ($\pm$ 46.2)	178.6 ($\pm$ 60.6)	0.040
Discharge therapy, n (%)				
High-dose high-intensity statins	153 (92.7)	102 (98.1)	51 (83.6)	<0.001
Ezetimibe	153 (92.1)	100 (96.2)	53 (86.9)	0.017
Combination: high-dose intensity statin and ezetimibe	145 (87.9)	98 (94.2)	47 (77.0)	0.002
Combination: statin and ezetimibe	147 (89.1)	99 (95.2)	48 (78.7)	0.001
Fibrate	2 (1.2)	1 (1.0)	1 (1.6)	ns
PCSK9 inhibitors	4 (2.4)	3 (2.9)	1 (1.6)	ns
Inclisiran	1 (0.6)	1 (1.0)	0	ns
Bempedoic acid	6 (3.6)	3 (2.9)	3 (4.9)	ns
Triple combination (Statin + ezetimibe + PCSK9 antibody/inclisiran/bempedoic acid)	8 (4.8)	7 (6.7)	1 (1.6)	ns

Bold values represent statistical significance.

Percentages are calculated from available data; denominators vary per variable.

ACS, acute coronary syndrome; BMI, body mass index; CVD, cerebrovascular disease; COPD, chronic obstructive pulmonary disease; IQR, interquartile range; LDL-C, low-density lipoprotein cholesterol; LLT, lipid-lowering therapy; non-HDL-C, non-high-density lipoprotein cholesterol; PAD, peripheral artery disease; PCSK9, proprotein convertase subtilisin/kexin type 9; SD, standard deviation.

after the index event), an additional 47 out of 57 patients (82.5%) had reached the target values. During the FU3, all five patients had reached the recommended target values (Figure 2). Altogether 93.8% were at goal within 8–12 weeks, and 100% of the study population were at goal within 12–16 weeks after the discharge for the index event.

Patients who did not reach the LDL-C target at FU1 had more frequent history of cerebrovascular disease (5.8% vs. 16.4%, $P = 0.026$) and more often reported statin intolerance (8.7% vs. 36.1%, $P < 0.001$) and were less frequently discharged on high-dose intensity statins and ezetimibe (94.2% vs. 77.0%, $P < 0.001$) (Table 1).

In total, dual therapy with a high-dose intensity statin and ezetimibe was sufficient to achieve target lipid levels in 98 (62.8%) of patients. Triple therapy consisted of a high-dose intensity statin, ezetimibe, and the addition of a third agent such as bempedoic acid (14/42 cases), or a PCSK9 inhibitor as evolocumab (14/42), alirocumab (4/42), or inclisiran (10/42), in total in 42 (26.9%) of cases (Figure 1). Rarely used regimens included monotherapy with a PCSK9 inhibitor (1.3%), dual therapy with a statin and bempedoic acid (0.6%), combinations of a

PCSK9 inhibitor with either ezetimibe (1.9%) or bempedoic acid (1.9%), or quadruple therapy (0.6%, statin + ezetimibe + bempedoic acid + evolocumab), respectively (Table 2).

Lipid levels over time

At baseline, the median LDL-C was 147.0 mg/dL (IQR 60.8) or 3.8 mmol/L. At the first follow-up, LDL-C had decreased to a median of 50.8 mg/dL (IQR 34.9), or 1.31 mmol/L. By FU2, the median LDL-C further decreased to 38.6 mg/dL (IQR 28.3), or 1.0 mmol/L, respectively (Table 3).

Adverse events

Of the 165 patients included in the study, 31 (18.8%) reported statin intolerance or statin-associated side effects. These adverse events included elevated CK ($n = 2$, 1.2%), elevated liver enzymes ($n = 6$, 3.6%), myalgia ($n = 21$, 12.7%), rhabdomyolysis ($n = 1$, 0.6%), and non-specific abdominal discomfort ($n = 1$, 0.6%), respectively.

Table 2 Therapy regimens needed to achieve the LDL-C target

	(n = 156)
Monotherapy with statin, n (%)	4 (2.6)
Monotherapy with ezetimibe, n (%)	2 (1.3)
Monotherapy with alirocumab, n (%)	2 (1.3)
Dual therapy: high-dose intensity statin + ezetimibe, n (%)	98 (62.8)
Statin + bempedoic acid, n (%)	1 (0.6)
Ezetimibe + PCSK9 inhibitor, n (%)	3 (1.9)
Evolocumab	2 (1.3)
Alirocumab	1 (0.6)
Inclisiran	0
PCSK9 inhibitor + bempedoic acid, n (%)	3 (1.0)
Evolocumab	
Triple therapy, n (%)	42 (26.9)
High-dose intensity statin + ezetimibe + bempedoic acid	14 (9.0)
High-dose intensity statin + ezetimibe + evolocumab	14 (9.0)
High-dose intensity statin + ezetimibe + alirocumab	4 (2.6)
High-dose intensity statin + ezetimibe + inclisiran	10 (6.4)
Quadruple therapy (statin + ezetimibe + bempedoic acid + evolocumab)	1 (0.6)

PCSK9, proprotein convertase subtilisin/kexin type 9.

Table 3 Cholesterol profile in patients during the follow-up

	Baseline (n = 165) Median (IQR)	Follow-up 1 (n = 165) Median (IQR)	Follow-up 2 (n = 57) Median (IQR)	P-value
LDL-C (mg/dL)	147.0 (60.8)	50.8 (34.9)	38.6 (28.3)	<0.001
Non-HDL-C (mg/dL)	177.0 (59.8)	74.0 (42.5)	61.0 (44.0)	<0.001

Bold values represent statistical significance.

IQR, interquartile range; LDL-C, low-density lipoprotein cholesterol; non-HDL-C, non-high-density lipoprotein cholesterol.

Discussion

General discussion

This study demonstrates that with a structured, stepwise lipid-lowering strategy initiated during hospitalization and continued in a dedicated outpatient setting, lipid target levels can be timely met in most patients at very high cardiovascular risk. Within 12 weeks, 93.8% of patients attending outpatient follow-up reached the targets recommended by the 2019 ESC/EAS guidelines. These results significantly surpass those observed in contemporary real-world studies such as DA VINCI and SANTORINI, which reported goal attainment rates of approximately 33% and 21%, respectively.^{7,8}

Our findings suggest that in-hospital initiation of a dual combination LLT (high-dose intensity statin plus ezetimibe) and timely up-titration to triple (or even quadruple) therapy are key components of successful LDL-C goal achievement. The vast majority of patients reached their targets with dual or triple therapy, with only a small minority requiring further intensification/adaptation. Importantly, the implementation of this protocol in a real-world setting highlights its practical feasibility and potential for broader adoption.

Successful lipid lowering in our cohort may be attributed to several factors. Firstly, the early initiation of LLT during hospitalization (the 'strike early' principle) ensured that all patients during follow-up already received dual LLT. Secondly, the rapid escalation protocol at follow-up visits (the 'strike-strong' principle) minimized therapeutic inertia, a known barrier to successful lipid control. Thirdly, the involvement of a specialized outpatient lipid ward likely enhanced adherence through patient education and close monitoring.

Our findings that the SES strategy¹⁴ may significantly contribute to meeting LDL/Non-HDL-C targets and downstream improvement of cardiovascular outcomes are supported by recent evidence.^{10–13} Patti

et al. published a retrospective analysis comparing three cohorts with varying ESC guideline LDL-C targets across different periods.¹⁰ At the first follow-up, 1-month post-discharge, the cohort with a target LDL-C of 70 mg/dL achieved a 55% goal attainment rate. In contrast, the 2nd cohort, treated under the updated 55 mg/dL target, achieved only 33%. Notably, in the third cohort, where the investigators implemented a 'strike early' approach combining early initiation of dual therapy, the goal attainment rate significantly improved to 83% for the 55 mg/dL target.¹⁰

Further support for this strategy comes from the EVOPACS trial, which demonstrated that early initiation of PCSK9 inhibitors in ACS patients resulted in LDL-C target achievement (<55 mg/dL or 1.4 mmol/L) in approximately 90% of cases.¹² Similarly, additional positive effects of this strategy have been shown by the PACMAN-AMI-trial investigating the addition of alirocumab right after acute myocardial infarction. In this study, patients receiving alirocumab demonstrated a significant reduction in coronary plaque volume, assessed via intravascular ultrasound, compared with placebo, confirming the favourable impact of PCSK9 inhibitors on plaque burden and composition.¹⁵ The HUYGENS study further confirmed these findings by examining the extent of plaque composition using optical coherence tomography in a group using the PCSK-9 inhibitor, evolocumab, compared with placebo in the cohorts of non-ST-elevation myocardial infarctions (NSTEMI). Patients treated with evolocumab exhibited greater plaque stabilization and regression compared with placebo, emphasizing the multifaceted benefits of aggressive LLT beyond LDL-C reduction alone.¹⁶

Important to note here is that follow-ups in the two above-mentioned studies have been performed only after 12 months and it was not clear how early plaque stabilization occurred. In a recent study from Japan (Kataoka et al.), it was suggested that plaque stabilization might occur much earlier in the time course of LLT. This study, with a population of only 30 participants, investigated the plaque evolution after early initiation of evolocumab using intravenous ultrasound. After only 7 weeks, a significant change in plaque composition was registered, which further supports the recommendation that optimal LLT should be initiated early to meet lipid targets.¹⁷

Randomized trials on early initiation of dual or triple LLT are currently ongoing, and studies such as VICTORION-INCEPTION,¹⁸ EVOLVE-MI,¹⁹ and the AMUNDSEN trial²⁰ will help definitively answer whether early adoption of PCSK9i-targeting improves clinical outcomes and thus refine guidelines for the management of lipid risk in this patient population at high cardiovascular risk.

Until then, observational registry data, such as a retrospective analysis from the SWEDEHEART registry suggesting greater absolute and relative MACE reduction with the early addition of ezetimibe to high-intensity statin therapy after ACS as compared

with a delayed initiation of combination therapy,²¹ may help guide an early adoption of more aggressive LLT initiation in higher-risk patients.²²

Strengths and limitations of the study

A major strength of this study lies in its real-world design, reflecting routine clinical practice within a tertiary care setting. Unlike randomized controlled trials that often include highly selected patient populations, this study captures the complexity and variability of patients encountered in everyday clinical work. The structured, guideline-based stepwise escalation approach—initiated during hospitalization and continued through dedicated lipid clinic follow-up—demonstrates the feasibility of implementing the 2019 ESC/EAS recommendations in a pragmatic setting. But, also, several limitations must be acknowledged. This study was non-randomized, and performed in a single, tertiary care centre. A comparator group receiving standard of care was not recruited, as well as there were not performed any further follow-ups for patients who did not show up to the appointment in the lipid clinic. Roughly, one-third of patients did not attend follow-up at the outpatient lipid clinic. Hence, patients who did not attend follow-up visits may differ systematically from those who did—e.g. in terms of medication adherence, risk profile, or health-seeking behaviour—which could impact the generalizability of our findings and our estimates may be biased.

Conclusion

This real-world investigation demonstrates that with a structured approach to LLT, beginning with a combination therapy of high-dose intensity statin and ezetimibe during hospitalization followed by an escalation after 4–6 weeks in a dedicated outpatient setting, 93.8% of patients attending outpatient follow-up reached ESC-recommended lipid goals within 12 weeks. These results significantly exceed goal attainment rates reported in large European registries and reinforce the clinical utility of a strike early and strong lipid-lowering strategy.

While observational data may promote early adoption of this strategy, further prospective, randomized studies are needed to confirm these results and to evaluate the long-term effect on hard cardiovascular outcomes of this SES approach.

Author contributions

Patrick Weltler (Data curation [lead], Investigation [lead], Project administration [equal], Resources [lead], Visualization [lead], Writing—original draft [lead], Writing—review & editing [equal]), Paul F. Harbich (Data curation [equal], Investigation [equal], Resources [equal]), Achim Burger (Conceptualization [equal], Investigation [equal], Methodology [equal]), Stephan Dobner (Methodology [equal], Supervision [equal], Writing—review & editing [equal]), Christoph C. Kaufmann (Investigation [equal], Methodology [equal]), Alexander Geppert (Conceptualization [supporting], Project administration [supporting], Supervision [supporting], Validation [supporting], Visualization [supporting]), Kurt Huber (Conceptualization [equal], Investigation [supporting], Methodology [supporting], Project administration [equal], Supervision [equal], Writing—review & editing [lead]), and Edita Pogran (Conceptualization [lead], Data curation [supporting], Formal analysis [lead], Methodology [lead], Supervision [equal], Validation [equal], Writing—original draft [supporting], Writing—review & editing [equal])

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Conflict of interest: None declared.

Data availability

The data of this study are available from the corresponding author upon reasonable request.

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